

Homogeneity of the water + ethanol + toluene azeotrope at 101.3 kPa

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Abstract

An all-glass, dynamic recirculating still equipped with an ultrasonic homogenizer has been used in the determination of the vapour–liquid equilibrium (VLE) and vapour–liquid–liquid equilibrium (VLLE). Consistent data for the ternary water + ethanol + toluene system are reported at 101.3 kPa at temperatures in the range of 347–357 K. These data have been compared with previously published VLLE data. The experimental data indicate that a ternary azeotrope is present at 347.60 K. It has been determined experimentally that the ternary azeotrope is homogeneous and not heterogeneous as is claimed in some of the studied references. The experimental results have been used to check the accuracy of the UNIFAC, UNIQUAC and NRTL models, proving that these models predict a bigger heterogeneous region than the experimental one.
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1. Introduction

One of the most frequently cited industrial examples of azeotropic distillation is ethanol dehydration using toluene as entrainer [1–4]. In fact, it is easy to find engineering companies that design plants to obtain anhydrous ethanol which name toluene as the most widely used entrainer in their projects.

Design of this industrial separation process suitable for the dehydration of ethanol by azeotropic distillation requires knowledge of the equilibrium data of the system. Various researchers have paid attention to the study of the water + ethanol + toluene ternary system and several papers have been published about the liquid–liquid equilibrium (LLE) at constant temperature [5–18] or at the boiling temperature [19,20].

However, vapour–liquid equilibrium (VLE) data at a constant pressure of 101.3 kPa have only been determined by three different authors [17,19,20] and no researchers have published vapour–liquid–liquid equilibrium (VLLE) data for this system. Knowledge of these data is essential to the design of an azeotropic distillation column to the dehydration of ethanol.

Regarding the type of ternary azeotrope water–ethanol–toluene is at atmospheric pressure, disparities exist among the data. Some authors classify it as a minimum-boiling heterogeneous azeotrope and others as a minimum-boiling homogeneous azeotrope. Note that although toluene could be used as an entrainer in the heterogeneous azeotropic distillation of ethanol and water, this does not necessarily imply that the water + ethanol + toluene ternary system contains a ternary heterogeneous azeotrope. The necessary requirement is that the vapour leaving the top of the azeotropic column splits into two liquid phases when it is condensed and subcooled.

Consulting azeotropic databases we found the following.

In the old azeotropic database of Horsley [21] where the type of azeotrope is not reported, two references have been encountered, one of Lecat [22] and the other from Union Carbide Chemicals Co. [23].

In the more complete azeotropic database of Gmehling et al. [24], in which the type of azeotrope is specified, the ternary azeotrope water–ethanol–toluene is classified as heterogeneous. Three different bibliographic references are cited in the database regarding this ternary azeotrope.

In the first, Young and Fortey [25], only the boiling temperature is given. In the second, Arzhanov et al. [19], the azeotrope is classified as heterogeneous. However, on the graphical repre-

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sentation of the LLE and VLE data shown in their paper, the azeotropic point is placed clearly outside of the heterogeneous region, delimited by the non-isothermal LLE curve, inside the homogeneous region. Finally Ricna et al. [18] state, without leaving room for doubt, that the azeotrope is homogeneous. They carried out this determination using a one meter long packed column containing a mixture with a composition close to that reported in literature [19].

Of course, none of these authors gives the composition of the two liquid phases into which the azeotrope would split at the bubble point. Therefore, the assertion of Gmehling et al. [24] that from these papers it can be deduced that the azeotrope is heterogeneous, is not true.

The most recent “DETERM” database [26] also classifies the azeotrope as heterogeneous. It includes three references: Arzhanov et al. [19] and Ricna et al. [18] mentioned above, and a new one, Borisova et al. [20]. The compositions of the two liquid phases in equilibrium with the vapour of the azeotrope are not shown either.

Nan and Tan [27] are the only authors who in a recent paper give these compositions. Curiously the composition and temperature of the azeotrope perfectly match those of Union Carbide Chemicals Co. [23]. But they report neither the experimental methodology used to determine the composition of the two liquid phases in equilibrium with the azeotropic vapour nor make reference to where these compositions have been obtained.

On the other hand, thermodynamic models as UNIFAC (Dortmund and Original), UNIQUAC and NRTL (with parameters taken from the literature (DECHEMA [28])), predict a ternary heterogeneous azeotrope. In fact, Gmehling [29] suggested toluene as a suitable entrainer for ethanol dehydration based on the UNIFAC (Dortmund) model, since it forms a ternary heterogeneous azeotrope.

Since the recent tendency is to consider that the water–ethanol–toluene system has a ternary heterogeneous azeotrope at atmospheric pressure [30], whereas older data indicates that it is homogeneous, it seems necessary to make a study of the VLLE of the system, which data are not available in the literature. The study reported in this article will permit distinction between homogeneity and heterogeneity for the ternary azeotrope and contribute to a better knowledge of the VLLE which is useful for the design of azeotropic distillation processes.

2. Experimental

2.1. Chemicals used

All reagents used were “for analysis” grade and obtained from Merck. The purity of ethanol and toluene is higher than 0.998 (mass fraction) and for 1-propanol higher than 0.995, so no further purification was needed. The 1-propanol was used as the internal standard for gas chromatography. The water content in mass% was determined by means of the Karl–Fischer titration method, and for toluene, ethanol and 1-propanol was around 0.003, 0.08 and 0.07%, respectively. The water used was bidistilled and purified using a Milli Q-Plus system.

2.2. Apparatus and procedures

An all-glass dynamic recirculating still with an ultrasonic homogenizer (Braun Labsonic P) coupled to a boiling flask was used to determine the VLLE data. This is a commercial device (Labodest model 602), assembled in Germany by Fischer Labor und Verfahrenstechnik and modified by Gomis et al. [31]. The use of ultrasound on the boiling flask causes emulsification of the two liquid phases throughout the still and thus prevents the fluctuations in temperature and flow rate of systems with two liquid phases. Visual observation indicates that the emulsified state is maintained throughout the apparatus during operation. The experimental procedure used is the same as that reported in previous studies [32–34]. Consequently, only the essential parts are reproduced here.

For VLE determinations, the apparatus was used without further modification since it allows for good mixing and separation of the vapour and liquid phases once they reach equilibrium.

A Pt-100 sensor was employed to measure the equilibrium temperatures. The probe was connected to a Cropico thermometer (model 3002) having an uncertainty of 0.006 K according to the calibration certificate (scale ITS 90 [35]). A Fischer M101 control system was used to measure and control the pressure and the heating power. The pressure in the still was 101.3 kPa, measured and controlled to an accuracy of 0.1 kPa. In order to guarantee the correct operation of the equipment, the boiling point of water was measured and checked with that in the bibliography.

Sampling was carried out by three different methods:

- Gaseous samples were injected into the GC through an UW Type, 6-port valve from Valco Instruments Co. The connecting tube walls were superheated with a resistance tape controlled by a potentiometer so that the vapour becomes unsaturated and condensation is avoided.
- For sampling of the liquid phase in the heterogeneous region, a small amount of the liquid coming from the separation chamber of the instrument was diverted to a tube using a solenoid valve. The test tube was placed in a thermostatic bath at a temperature equal to the boiling point of the mixture. In this way, the dispersed liquid phases enter this tube and separate into two layers at their bubble point. A sample of each layer was taken and placed in a vial with a small amount of 1-propanol as an internal standard. Between each equilibrium determination the drops remaining in the tube, which feeds the sample test tube, were removed by vacuum and external heating.
- In the homogeneous region, samples were withdrawn from the liquid coming from the separator chamber using a syringe and put into a vial with a small amount of internal standard.

All analytical work was carried out by gas chromatography on a Shimadzu GC-14B coupled with a personal computer through the Shimadzu CLASS-VP Chromatography Data System. Separation of the components was obtained in a 2 m × 3 mm column

packed with Porapak Q 80/100. The oven temperature was 453 K and the helium flow rate was 40 mL/min.

Detection was carried out by different techniques depending on the composition of the samples: TCD (thermal conductivity detector), for organic and aqueous samples (analysis for water, ethanol and toluene) and FID (flame ionization detector) for aqueous samples (analysis for ethanol and toluene). The temperature of the detector was 473 K and the current for the TCD was 100 mA. Water in the organic phase was also determined using the Karl–Fischer titration method.

An internal standard was used to obtain quantitative results from the analysis of the liquid phases. For this reason, 1-propanol, which is completely miscible in water, ethanol and toluene, was also added to the sample vials. Furthermore, the addition of the standard prevents phase split when changing the temperature after the separation of the phases.

The accuracy of the mole fraction measurements was estimated at ± 0.002 for all the compounds except for the water in the liquid organic phase and toluene in the aqueous phase, where the accuracy was approximately ± 0.005 .

In order to check if the azeotropes found in the literature were homogeneous, samples of known composition by weight were prepared in a hermetically sealed test tube. The tube, which contained two liquid phases at room temperature, was placed in a thermostatic bath at the boiling temperature. At the same time, it was stirred vigorously and then left to stand for at least 2 h to allow it to reach equilibrium. The presence of one or two liquid phases was then observed.

Once this experiment was done, the upper critical solution temperature of the samples was determined. The test tube containing the prepared sample was placed in a thermostatic bath at room temperature and it was stirred continuously. The temperature of the bath was then slowly increased until only one liquid phase remained. The process was repeated starting at temperatures for which the mixture was homogeneous and subsequently decreasing the temperature until a heterogeneous mixture was obtained. The differences between both the temperatures measured were always less than 0.5 °C.

3. Results and discussion

The experimental VLLE data for the heterogeneous water–toluene binary azeotrope (BIN) and the corresponding ternary system studied are reported in Table 1. All these data are shown in Fig. 1, which on the graphical ternary diagram represents the tie lines, the vapour line and the non-isothermal binodal curve. The last one represents the projection of the intersection of the single liquid–liquid envelope with the VLE surface onto the ternary composition diagram. It may look like an isothermal liquid–liquid envelope but it is not.

The isobaric data have been compared with those obtained by other authors. As can be appreciated in Fig. 2, the heterogeneous region delimited by the non-isothermal binodal curve is very similar to that determined by Borisova et al. [20]. However, the heterogeneous region is larger than that obtained by Arzhanov et al. [19]. The slope of the tie lines in the bottom part of the ternary diagram is similar

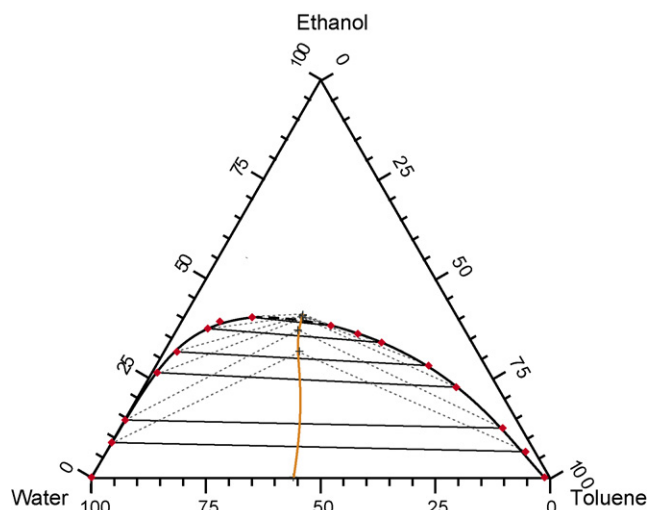


Fig. 1. VLLE (mol%) diagram for the water + ethanol + toluene ternary system at 101.3 kPa: (♦) liquid phase; (+) vapour phase.

in all cases. However, in the top part there are more differences.

VLE data for the homogeneous region are shown in Table 2 and are graphically presented in Fig. 3. The ternary VLE and VLLE experimental data have been tested by the point-to-point L–W Wisniak [36] method and were found to be thermodynamically consistent. Calculations involved in the test were done using the program PRO-VLE 2.0 [37]. All the L/W values are between 0.98 and 1.00. The test did not reveal any significant inconsistency in the data. Vapour pressures for the three components were calculated using the Antoine equation, with parameters A_i , B_i , and C_i taken from literature [38] and shown in Table 3.

It is worth noting that in the middle of the diagram in Fig. 3 a liquid point with a vapour in equilibrium very close to it can be seen. Furthermore, it has the lowest boiling temperature of the data determined in this study, 347.60 K. On the other hand, this

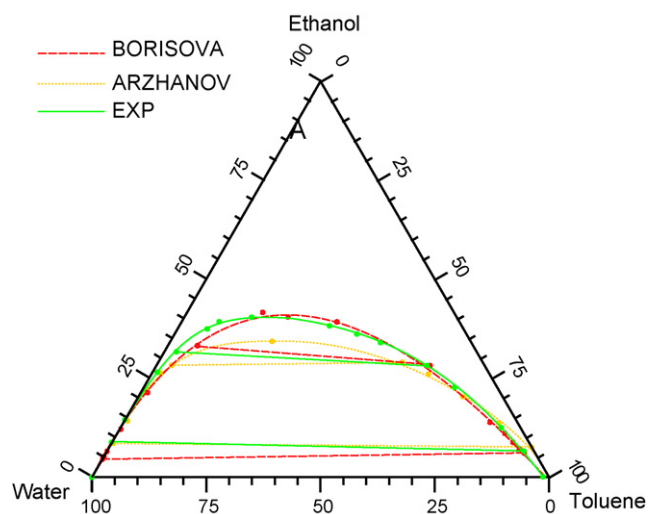


Fig. 2. Comparison of the non-isothermal binodal curve and the tie lines obtained by various authors.

Table 1

Vapour–liquid–liquid equilibrium data (mole fraction) for the water (1) + ethanol (2) + toluene (3) ternary system at 101.3 kPa

	Organic phase			Aqueous phase			Vapour phase			T_b (K)
	x_1	x_2	x_3	x_1	x_2	x_3	y_1	y_2	y_3	
BIN	0.012	0.000	0.988	0.9997	0.0000	0.0003	0.561	0.000	0.439	357.34
1	0.022	0.064	0.914	0.912	0.087	0.001	0.389	0.316	0.295	350.04
2	0.042	0.123	0.835	0.855	0.144	0.001	0.365	0.370	0.265	348.63
3	0.092	0.226	0.682	0.725	0.263	0.012	0.338	0.401	0.261	347.91
4	0.125	0.280	0.595	0.656	0.315	0.029	0.337	0.405	0.258	347.70
5	0.199	0.338	0.463	0.560	0.373	0.067	0.335	0.408	0.257	347.67
6	0.239	0.360	0.401	0.525	0.391	0.084	0.334	0.409	0.257	347.64
7	0.288	0.381	0.331	0.449	0.402	0.149	0.334	0.410	0.256	347.64

Table 2

Vapour–liquid equilibrium data (mole fraction) for the water (1) + ethanol (2) + toluene (3) ternary system at 101.3 kPa

T_b (K)	Liquid phase			Vapour phase		
	x_1	x_2	x_3	y_1	y_2	y_3
351.47	0.426	0.562	0.012	0.334	0.607	0.059
350.78	0.146	0.833	0.021	0.148	0.805	0.047
350.24	0.228	0.736	0.036	0.207	0.704	0.089
350.20	0.412	0.558	0.030	0.286	0.586	0.128
349.17	0.256	0.664	0.080	0.227	0.609	0.164
349.76	0.550	0.422	0.028	0.300	0.509	0.191
348.72	0.444	0.490	0.066	0.298	0.492	0.210
348.40	0.270	0.594	0.136	0.244	0.535	0.221
351.07	0.653	0.336	0.011	0.322	0.517	0.161
348.21	0.417	0.483	0.100	0.291	0.466	0.243
348.39	0.198	0.604	0.198	0.222	0.552	0.226
348.09	0.268	0.533	0.200	0.266	0.486	0.248
348.29	0.157	0.543	0.300	0.230	0.524	0.245
348.25	0.152	0.501	0.347	0.237	0.508	0.255
348.04	0.153	0.402	0.445	0.280	0.452	0.268
348.10	0.256	0.562	0.183	0.244	0.506	0.250
348.54	0.149	0.626	0.225	0.192	0.572	0.236
349.09	0.080	0.663	0.257	0.139	0.631	0.229
349.02	0.077	0.585	0.338	0.141	0.622	0.237
348.96	0.081	0.533	0.386	0.159	0.589	0.252
348.72	0.081	0.430	0.489	0.195	0.541	0.264
348.51	0.457	0.453	0.089	0.320	0.470	0.210
348.56	0.448	0.471	0.082	0.315	0.482	0.203
348.61	0.487	0.446	0.066	0.320	0.482	0.198
347.62	0.383	0.433	0.184	0.327	0.425	0.248
347.66	0.434	0.395	0.170	0.333	0.413	0.254
347.61	0.347	0.391	0.262	0.332	0.412	0.256

point is also very close to the non-isothermal binodal curve. The minimum temperature of the vapour line is 347.64 K and the temperature of the VLE data closest to it is 347.65 K. The proximity between the vapour and the liquid and the lowest temperature suggests that the composition of the ternary azeotrope for this

system cannot differ significantly from that of its homogeneous vapour. The azeotropic composition cannot be determined by numerical interpolation as in previous studies [32–34], since the ternary azeotrope is homogeneous. For this reason, the composition and temperature of the ternary azeotrope was established as $x_1 = 0.332$, $x_2 = 0.412$, and $x_3 = 0.256$, and 347.60 K, respectively. These data do not differ significantly from those found in literature [18,21,24], as can be seen in Table 4.

The given experimental data for the azeotrope at atmospheric pressure correspond to a minimum-boiling homogeneous azeotrope, contrary to what is claimed in the most recent compilations (Gmehling et al. [24] and DETHERM [26]), where the water–ethanol–toluene system at atmospheric pressures is considered to have a ternary heterogeneous azeotrope. In order

Table 3

Antoine equation parameters for the pure substances^a

Compound	A	B	C	Temperature range (°C)
Water	7.96681	1668.21	228.0	60/150
Ethanol	8.04494	1554.3	222.65	n.a.
Toluene	6.95334	1343.943	219.377	35/111

^a Antoine eq. : $\log(P) = A - B/[T + C]$, with: P (mmHg) and T (°C).

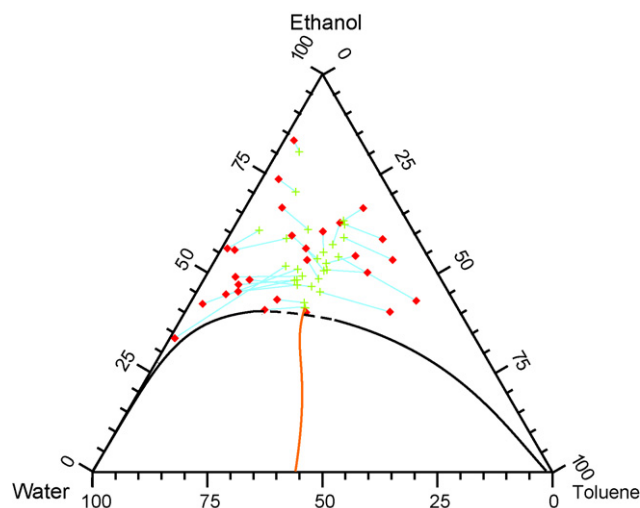


Fig. 3. VLE (mol%) diagram for the water + ethanol + toluene ternary system at 101.3 kPa: (♦) liquid phase; (+) vapour phase.

to confirm our assertion on the one hand, the azeotropic mixtures of the various authors were placed in a thermostatic bath at the azeotropic temperature (347.55 K). In all cases the azeotropes were found to be homogeneous. On the other hand, the upper critical solution temperature of each one of the mixtures was determined. The values are presented in Table 4 and all of them are lower than the azeotropic temperature. In conclusion, the azeotrope is homogeneous.

On the other hand, it is true that all the azeotropic compositions of Table 4 produce heterogeneous mixtures at ambient temperature, causing confusion about the type of azeotrope, as is also caused by the fact that thermodynamical models as UNIFAC (Dortmund and Original), UNIQUAC and NRTL predict a ternary heterogeneous azeotrope. Fig. 4 shows the graphical representation of two different experimental tie lines compared with those obtained using UNIQUAC and NRTL. Results for the tie lines obtained by UNIFAC (Original) are also shown. All the calculations have been made using CHEMCAD V [39]. The slope of the tie lines in the bottom part of the diagram, calculated using NRTL, UNIQUAC and UNIFAC, is steeper than the experimental one. This behavior becomes more pronounced in the top part of the diagram. In all cases, it can be observed that the experimental heterogeneous region is much smaller than the calculated one which even contains some homogeneous experimental points. Hence, the use of these models may lead to

Table 4

Ternary azeotrope for the water (1)–ethanol (2)–toluene (3) system (mole fractions) at 101.3 kPa as obtained by various authors

x_1	x_2	x_3	T_b (K)	T_{UCST} (K)	
0.332	0.412	0.256	347.61	324.4	Experimental
0.326	0.432	0.242	347.55	308.6	Arzhanov et al. [19]
0.329	0.397	0.274	347.55	331.8	Union Carbide Co. [23]
0.304	0.436	0.260	347.55	308.8	Ricna et al. [18]
0.326	0.395	0.279	347.55	333.1	Borisova et al. [20]
			347.70		Young and Fortey [25]

T_b , azeotropic temperature; T_{UCST} , upper critical solubility temperature.

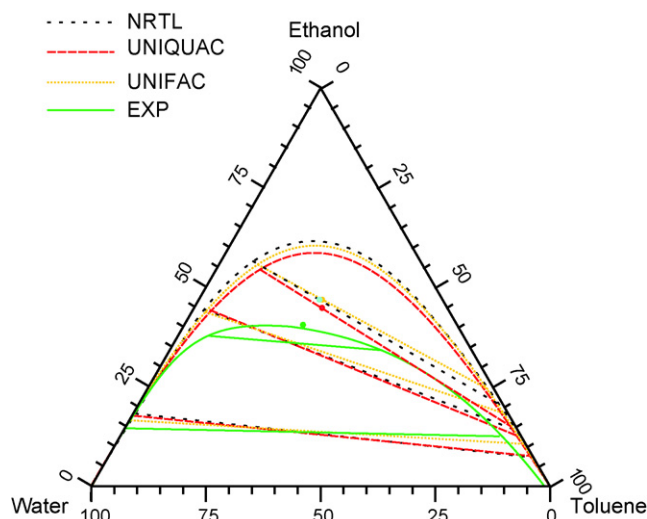


Fig. 4. Comparison of the non-isothermal binodal curve and some tie lines, including the azeotropic tie line, calculated with UNIFAC and with the parameters obtained with NRTL and UNIQUAC.

inaccurate predictions for the equilibrium and important differences in the simulation results. In fact, these models predict a heterogeneous azeotrope that can lead to erroneous simulations of the ethanol dehydration process [40]. In Fig. 4 are included the azeotrope composition and those of the two liquid phases in equilibrium with it. The composition of this azeotrope goes out of the experimental heterogeneous region in all cases.

4. Conclusions

The VLE and VLLE data obtained are consistent with data published previously. These data point to the existence of a ternary azeotrope.

Despite the fact that various bibliographic sources classify the ternary azeotrope as heterogeneous, it has been experimentally proven that the water–ethanol–toluene azeotrope is homogeneous.

VLLE data obtained by means of different activity-coefficient models have been compared with those which are experimental. These models always predict a heterogeneous region bigger than the experimental one. For this reason some experimental homogeneous mixtures are predicted as heterogeneous, including the ternary azeotrope.

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